GeoPops demonstration: An open-source, adaptable framework for agent-based modeling on synthetic populations

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Definition

GeoPops[1] is an open-source tool for generating geographically and demographically realistic synthetic populations for any US Census area from publicly available data with associated networks for home, school, workplace, and group quarters (GQ) (dormitories, nursing homes, etc.). GeoPops has the same underlying methodology as GREASYPOP-CO[2] with the following changes:

- Wrapped in convenient Python package
- Can be pip installed
- Compatibility with Census data beyond 2019 (still developing)
- Automated data downloading
- Users can adjust all config parameters from the front-end
- Compatibility with the agent-based modeling software Starsim[3]

github.com/ACCIDDA/GeoPops

pypi.org/project/geopops

Methodology

Step 1: Generate individuals and households

Attributes	Combinatorial Optimization (CO)
• Age • Gender • Race/ethnicity • Worker status • Student status • School grade • Commuter status • Income category • Workplace category • Home CBG • School CBG • Workplace CBG	Public Use Microdata Samples (PUMS) → CO samples households from PUMS until a set is found that reasonably matches ACS target variables → American Community Survey (ACS)

Step 2: School and Workplace Assignment

	Methods	Data	Data Sources
Schools	• Students assigned to closest school that offers required grade • Teachers assigned to school by selecting teachers from nearby CBGs	Student and staff counts for each grade level in public schools	National Center for Education Statistics (NCES)
Workplaces	• Iterative proportional fitting (IPF) to make origin-destination commute matrix • IPF to make industry by destination commute matrix for each origin CBG • Includes outside workers	Origin-destination and workplace category matrices	Longitudinal Employer-Household Dynamics (LEHD)

Step 3: Network Generation

Network	Algorithm	Parameters
Household	All agents within a household are fully connected	• N/A
School	Stochastic Block Model within each school	• Assortativity: α=0.9 • Mean degree: K=12 • Blocks are grade levels (Pre-k – 12)
Workplace	Stochastic Block Model within each workplace	• Assortativity: α=0.9 • Mean degree: K=8 • Blocks are income levels (<40k or ≥ 40k)
Group quarters	Watts-Strogatz Model within each GQ facility	• Mean degree: K=12 • Rewiring probability: β=0.25

Usage

Code

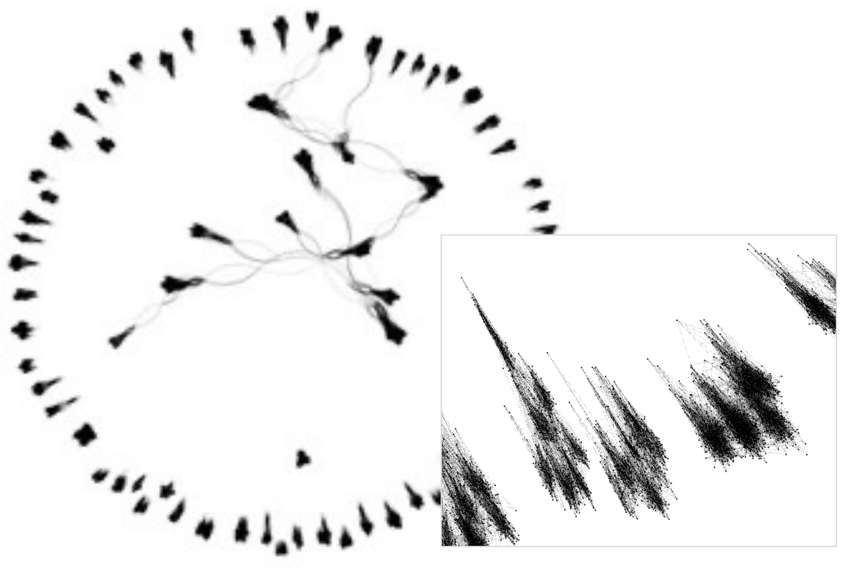
Example for Howard County, MD

```
import geopops

pars = {'path': 'YOUR_OUTPUT_DIR',     # Where you want outputs
        'census_api_key': 'YOUR_KEY',  # Your Census API key
        'main_year': 2019,             # Main year
        'geos': ['24027'],             # State/county FIPS of main geo
        'commute_states': ['24'],      # Commute to/from states
        'use_pums': ['24']}            # PUMS states to download

geopops.WriteConfig(**pars) # Set parameters
geopops.DownloadData()      # Download raw data files
geopops.ProcessData()       # Preprocessing for next steps
geopops.RunJulia()          # CO, network generation, export to csv
geopops.ForStarsim()        # Format for Starsim
```

Outputs

- Original files from each data source
- csv of agents and attributes
- csv of homes with locations (CBG level)
- csv of schools with locations and students and teachers by grade
- csv of workplaces with locations and workers by category
- csv of GQ facilities with locations and residents and workers by type (e.g., dormitory vs long-term care facility)
- csv of outside workers (people who commute into area for work)
- Files in matrix and csv format for each network
- Starsim compatible files for agents and networks

A



B

p1	p2	beta
748	78	1
984	883	1
1702	115	1
1709	1521	1
1875	651	1

Visualization and table snippet of the school network. (A) Nodes are students and teachers clustered in grades within schools. (B) Same graph in Starsim tabular format where beta is the weight of the connection between person one and person two.

Example of Howard County, MD*

Components passed into Starsim to model disease transmission

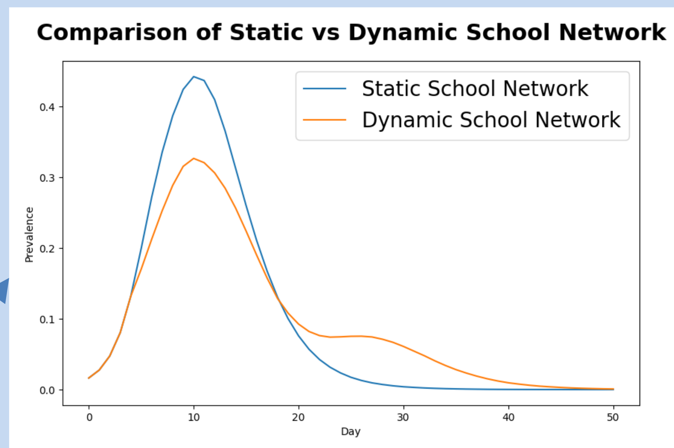
GeoPops agents

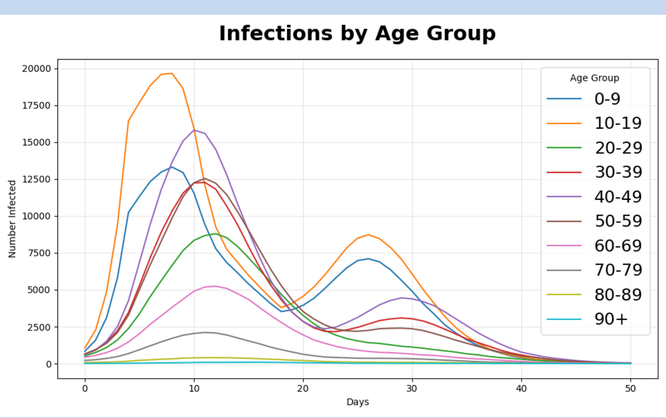
GeoPops networks

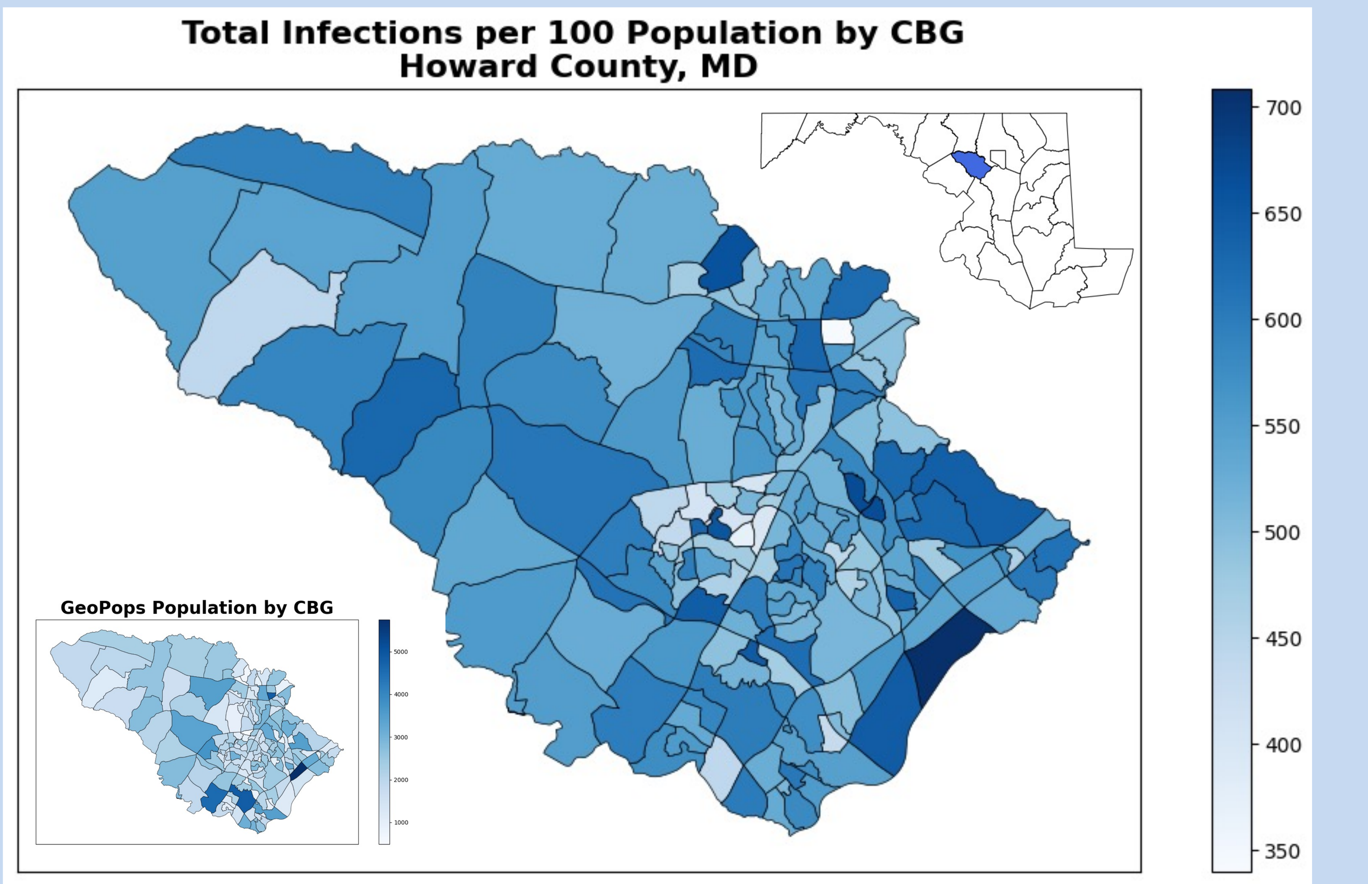
SIR model
- Upper respiratory disease
- Infection duration ~7 days
- Transmission coefficient ~0.1
- 50-day simulation

Feedback module
- Removes edges of school network when prevalence > 0.1

Subgroup Tracking
- Infections by age and CBG







*This example reflects a simple application with a generic upper respiratory infection.

References

Hamilton, Alisa, Sasha Tulchinsky, Gary Lin, Cliff Kerr, Eili Klein, and Lauren Gardner. 2025. GeoPops: Geographically Realistic Synthetic Population Generator. Johns Hopkins University and One Health Trust. https://github.com/ACCIDDA/GeoPops.

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Kerr, Cliff, Robyn Stuart, Romesh Abeysuriya, et al. 2024. "Starsim: A Fast, Flexible Toolbox for Agent-Based Modeling of Health and Disease." https://starsim.org/.

github.com/ACCIDDA/GeoPops/tutorials/MIDAS

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